

67. MIPI CSI Interface

67.1 Overview

The MIPI CSI-2 is a MIPI Camera Serial Interface 2 receiver module.

This module supports MIPI CSI-2 V3.0 and MIPI D-PHY V2.5 (80 Mbps to 720 Mbps/lane and up to 2 lanes).

The image signal received by the MIPI-CSI2 is output to the Video input module.

In this section, “vclk” is used to refer to PCLKA.

67.2 Features

Item	Specifications
Compliant specification	MIPI Alliance Specification for Camera Serial Interface 2 (CSI-2) SM Version 3.0
	MIPI Alliance Specification for D-PHY Version 2.5
CSI-2 support functions	Support Data Descrambling
	Generic Short Packet FIFO with the depth of 16
	Support Latency Reduction and Transport Efficiency (LRTE) - Support D-PHY EPD option 2 - Support variable-length spacer detection in D-PHY EPD option 2 - Support End-of-Transmission Short Packet (EoTp) detection
	ECC 1-bit error correction and 2-bit error detection in packet header
	CRC check for payload data
	Receiver Error Detection and Report - D-PHY Level errors - Packet Level errors - Protocol Decoding Level errors
CSI-2 non-support functions*1	Connection to C-PHY
	Unified Serial Link (USL)
	Smart Region of Interest (SROI)
	Alternate Low Power (ALP) mode
	Fast Lane Turnaround
	D-PHY EPD option 1

Note 1. Unsupported functions among the main functions specified in CSI-2 Ver. 3.0 and D-PHY Ver. 2.5.

For the block diagram and I/O pins, see [Figure 62.1](#) and [Table 62.1](#).

67.3 Register Descriptions

Access the following registers only in units of 32 bits.

67.3.1 MCG : Module Configuration Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	GSNM							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	SDLN				—	—	—	—	VER			
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	VER	Version of this IP Indicate the version of this core.	R
7:4	—	These bits are read as 0.	R
11:8	SDLN	Number of Supported Data Lanes Indicate the maximum number of supported data lanes. 0x2: Operable with 2 lanes or 1 lane	R
15:12	—	These bits are read as 0.	R
23:16	GSNM	Number of Generic Short Packet FIFO Indicate the number of stages of the generic short packet FIFO (16 stages).	R
31:24	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

67.3.2 MCT0 : Module Control Register 0

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	LFSR EN	ECCV 13	—	—	—	GRMD	RVMD	—	EDMD	ZLMD
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VDLN			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
3:0	VDLN	Number of Valid Data Lanes Set the number of used data lanes. Change this field when MCT3.RXEN = 0. A value equal to or less than the MCG.SDLN value can be specified. 0x1: Operation with one lane 0x2: Operation with two lanes Others: Setting prohibited	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
16	ZLMD	Zero Length Long Packet Output Mode Sets whether to output a long packet whose word count is 0 for the Video Pixel interface. 0: Output 1: No output	R/W
17	EDMD	ErrframeData Notification Mode Notifies ErrFrameData when an ECC 2-bit error or a packet of less than 4 bytes is received between FS and FE. 0: Does not notify ErrFrameData 1: Notifies ErrFrameData	R/W
18	—	This bit is read as 0. The write value should be 0.	R/W
19	RVMD	Reserved Packet Reception Mode Reception operation mode for packets with a data type of 0x39 to 0x3E, which is reserved in the CSI-2 standard "Table 10 Data Type Classes". 0: Discards the data (notifies an ErrID) 1: Receives data as a long packet (outputs data to Video Pixel interface when the corresponding bit in DTEH is 1, or discards the data and notifies an ErrID when the corresponding bit in DTEH is 0).	R/W
20	GRMD	Generic CSI-2 Rule Mode Mode to output all data output to Video Pixel interface according to CSI-2 standard Generic CSI-2 rule. Set to 1 for this LSI.	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
24	ECCV13	ECC Check CSI-2 Ver 1.3 Mode CSI-2 specification-compliant mode of ECC checking. It is prohibited to set this bit to 1 when receiving a packet whose virtual channel is from 4 to 15. 0: 26 bits to be checked for ECC 1: 24 bits to be checked for ECC	R/W
25	LFSREN	LFSR Enable Mode Enables or disables descrambling. 0: Disables descrambling 1: Enables descrambling	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.3.3 MCT2 : Module Control Register 2

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	FRRSKW								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	FRRCLK								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	FRRCLK	Frequency clock rate to determine packet reception end (internal setting parameter) Set the following value using the ROUNDDOWN function that rounds down the decimal places. $(\text{ROUNDDOWN}((1.5 \times \text{vclk frequency}/\text{hsclock frequency}), 0) + 1)$ Example: Set 3 when the vclk frequency is 125 MHz and the hsclock frequency is 90 MHz. $\text{ROUNDDOWN}((1.5 \times 125/90), 0) + 1 = 3$	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
24:16	FRRSKW	Frequency clock rate to adjust data lane skew (internal setting parameter) Set the following value using the ROUNDDOWN function that rounds down the decimal places. $(\text{ROUNDDOWN}((3 \times \text{vclk frequency}/\text{hsclock frequency}), 0) + 1)$ Example: Set 5 when the vclk frequency is 125 MHz and the hsclock frequency is 90 MHz. $\text{ROUNDDOWN}((3 \times 125/90), 0) + 1 = 5$	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.3.4 MCT3 : Module Control Register 3

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x01C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RXEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RXEN	RX (reception) Enable Enables data reception in the PPI. 0: Disables reception 1: Enables reception	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.3.5 RTCT : Reset Control Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x028

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VSRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VSRST	Video Pixel Interface Software Reset Writing 1 to this bit resets the internal LINK signal by a software reset.	W
31:1	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

67.3.6 RTST : Reset Status Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x02C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VSRSTS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VSRSTS	Video Pixel Interface Software Reset Status Indicates the execution status of the software reset generated by RTCT.VSRST. 0: Does not indicate status during a reset 1: Indicates status during a reset	R
31:1	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

67.3.7 EPCT : EPD Option Control Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EPDEN	SSP														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EPDOP	SLP														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	SLP	Long Packet Spacers Specify the number of spacers/lanes inserted after a long packet. This field is valid when EPDEN = 1. For EPD Option 2 variable length spacer selection (EPDOP = 1 and EMCT.VLSIEN not equal 00b), the number of spacers equal to or greater than the value of this field is inserted. When the fixed-length spacer is selected for EPD Option 2 (EPDOP = 1 and EMCT.VLSIEN = 00b), the same number of spacers as this field value is inserted. When EPDEN = 1, the minimum number of spacers is the value described in the SSP field. Setting a value less than the minimum value is prohibited.	R/W

Bit	Symbol	Function	R/W
15	EPDOP	EPD Option Select Selects an EPD option. Set this bit to 1. 0: D-PHY EPD Option 1 (0 cannot be set when EPDEN = 1) 1: D-PHY EPD Option 2	R/W
30:16	SSP	EPD Short Packet Spacers Specify the number of spacers/lanes inserted after a short packet. This field is valid when EPDEN = 1. For EPD Option 2 variable length spacer selection (EPDOP = 1 and EMCT.VLSIEN not equal 00b), the number of spacers equal to or greater than the value of this field is inserted. When the fixed-length spacer is selected for EPD Option 2 (EPDOP = 1 and EMCT.VLSIEN = 00b), the same number of spacers as this field value is inserted. Table 67.1 shows the specifiable minimum number of spacers when EPDEN = 1. Setting a value less than the minimum value is prohibited.	R/W
31	EPDEN	Enable EPD Operation Enables the EPD. 0: Disables the EPD 1: Enables the EPD	R/W

Note: S-TYPE-3, P-TYPE-3

The EPCT register supports the TX_REG_CSI_EPD_EN_SSP register and the TX_REG_CSI_EPD_OP_SLP register that are defined in Section 9.11.1.2.3, D-PHY EPD Specifications (for EPD Options 1 and 2) in the MIPI CSI-2 Specification.

Set the same value as the one set in this register in the CSI-2 transmitter.

Table 67.1 Specifiable minimum number of spacers when EPDEN = 1

EPDOP	Minimum number of spacers						
1	The larger value of the following two values: - Minimum number of spacers from the D-PHY setting - Minimum number of spacers from the number of used lanes						
	Minimum number of spacers from the D-PHY setting The minimum value can be found using the ROUNDUP function that rounds up the decimal places. ROUNDUP (((105/U[ns] + 20) / 8), 0)						
	Minimum number of spacers from the number of used lanes <table> <tr> <td>Number of used lanes</td><td>Minimum number of spacers</td></tr> <tr> <td>1 lane</td><td>64</td></tr> <tr> <td>2 lanes</td><td>32</td></tr> </table>	Number of used lanes	Minimum number of spacers	1 lane	64	2 lanes	32
Number of used lanes	Minimum number of spacers						
1 lane	64						
2 lanes	32						

67.3.8 EMCT : EPD Misc Option Control Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x044

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	EOTP EN	VLSIEN	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
5:4	VLSIEN	Enable Variable Length Spacer Insertions Enables variable length spacer insertion when EPD Option 2 is enabled. This field is valid when EPD Option 2 is enabled (EPCT.EPDEN = 1 and EPDOP = 1). When EPD Option 2 is disabled, use the default value (0x0). Set the following values according to the number of lanes used. 1 lane: 00b or 11b (01b and 10b are prohibited) 2 lanes: 00b or 10b or 11b (01b is prohibited) 0 0: Variable length spacer invalid (spacer number is fixed length) 0 1: Variable length spacer valid (spacer number is 1 × n/lane) 1 0: Variable length spacer valid (spacer number is 2 × n/lane) 1 1: Variable length spacer valid (spacer number is 4 × n/lane)	R/W
6	EOTPEN	Enable EOTP Enables EOTP insertion when EPD Option 2 is enabled. This field is valid when EPD Option 2 is enabled (EPCT.EPDEN = 1 and EPDOP = 1). When EPD Option 2 is disabled, use the default value (0). If variable length spacer is supported (VLSIEN not equal 00b), be sure to set this field to 1. 0: No EOTP 1: With EOTP	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The EMCT register supports the TX_REG_CSI_EPD_MISC_OPTIONS register that is defined in Section 9.11.1.2.3, D-PHY EPD Specifications (for EPD Options 1 and 2) in the MIPI CSI-2 Specification.

Set the same value as the one set in this register in the CSI-2 transmitter.

67.3.9 MIST : Module Interrupt Status Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x050

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	VC15S	VC14S	VC13S	VC12S	VC11S	VC10S	VC9S	VC8S	VC7S	VC6S	VC5S	VC4S	VC3S	VC2S	VC1S	VC0S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	RXS	GSTS	PMS	—	—	—	—	—	—	DL1S	DL0S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DL0S	Interrupt status of data lane 0 When this bit is set to 1, an interrupt source associated with data lane 0 exists. Check the DLST0 register for details of the interrupt source.	R
1	DL1S	Interrupt status of data lane 1 When this bit is set to 1, an interrupt source associated with data lane 1 exists. Check the DLST1 register for details of the interrupt source.	R
7:2	—	These bits are read as 0.	R
8	PMS	Interrupt status of power management When this bit is set to 1, an interrupt source associated with the power management exists. Check the PMST register for details of the interrupt source.	R
9	GSTS	Interrupt status of generic short packet When this bit is set to 1, an interrupt source associated with the reception of generic short packets exists. Check the GSST register for details of the interrupt source.	R
10	RXS	Interrupt status of RX (reception) When this bit is set to 1, an interrupt source associated with the reception in general exists. Check the RXST register for details of the interrupt source.	R

Bit	Symbol	Function	R/W
15:11	—	These bits are read as 0.	R
16	VC0S	Interrupt status of virtual channel 0 When this bit is set to 1, an interrupt source associated with virtual channel 0 exists. Check the VCST0 register for details of the interrupt source.	R
17	VC1S	Interrupt status of virtual channel 1 When this bit is set to 1, an interrupt source associated with virtual channel 1 exists. Check the VCST1 register for details of the interrupt source.	R
18	VC2S	Interrupt status of virtual channel 2 When this bit is set to 1, an interrupt source associated with virtual channel 2 exists. Check the VCST2 register for details of the interrupt source.	R
19	VC3S	Interrupt status of virtual channel 3 When this bit is set to 1, an interrupt source associated with virtual channel 3 exists. Check the VCST3 register for details of the interrupt source.	R
20	VC4S	Interrupt status of virtual channel 4 When this bit is set to 1, an interrupt source associated with virtual channel 4 exists. Check the VCST4 register for details of the interrupt source.	R
21	VC5S	Interrupt status of virtual channel 5 When this bit is set to 1, an interrupt source associated with virtual channel 5 exists. Check the VCST5 register for details of the interrupt source.	R
22	VC6S	Interrupt status of virtual channel 6 When this bit is set to 1, an interrupt source associated with virtual channel 6 exists. Check the VCST6 register for details of the interrupt source.	R
23	VC7S	Interrupt status of virtual channel 7 When this bit is set to 1, an interrupt source associated with virtual channel 7 exists. Check the VCST7 register for details of the interrupt source.	R
24	VC8S	Interrupt status of virtual channel 8 When this bit is set to 1, an interrupt source associated with virtual channel 8 exists. Check the VCST8 register for details of the interrupt source.	R
25	VC9S	Interrupt status of virtual channel 9 When this bit is set to 1, an interrupt source associated with virtual channel 9 exists. Check the VCST9 register for details of the interrupt source.	R
26	VC10S	Interrupt status of virtual channel 10 When this bit is set to 1, an interrupt source associated with virtual channel 10 exists. Check the VCST10 register for details of the interrupt source.	R
27	VC11S	Interrupt status of virtual channel 11 When this bit is set to 1, an interrupt source associated with virtual channel 11 exists. Check the VCST11 register for details of the interrupt source.	R
28	VC12S	Interrupt status of virtual channel 12 When this bit is set to 1, an interrupt source associated with virtual channel 12 exists. Check the VCST12 register for details of the interrupt source.	R
29	VC13S	Interrupt status of virtual channel 13 When this bit is set to 1, an interrupt source associated with virtual channel 13 exists. Check the VCST13 register for details of the interrupt source.	R
30	VC14S	Interrupt status of virtual channel 14 When this bit is set to 1, an interrupt source associated with virtual channel 14 exists. Check the VCST14 register for details of the interrupt source.	R
31	VC15S	Interrupt status of virtual channel 15 When this bit is set to 1, an interrupt source associated with virtual channel 15 exists. Check the VCST15 register for details of the interrupt source.	R

Note: S-TYPE-3, P-TYPE-3

67.3.10 DTEL : Receive Data Type Enable Low Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x060

Bit position: 31

0

Bit field:

DTEN[31:0]

Value after reset: 0 1 1 1 1

Bit	Symbol	Function	R/W
31:0	DTEN[31:0]	Data Type Enable (DT = 0x00 to 0x1F) Receive data of a packet whose data type is one of the following. After multiple data types are selected, only one data type is selected by the DT[5:0] bits in the CSI_IFMD register in the Video Input module. See Table 67.2 . 0: Does not receive data 1: Receives data	R/W

Note: S-TYPE-3, P-TYPE-3

Table 67.2 Data type selected by the DT[5:0] bits in the CSI_IFMD register in the Video Input module after multiple data types are selected for DTEN[31:0]

Bit	Data type	Attribute	Comment
31	YUV422 10-bit	R/W	When this bit is set to 1, data is output to the Video Pixel interface.
30	YUV422 8-bit	R/W	
29:16	Reserved (embedded 8-bit non image data)	R/W	Fixed to 0 because this function is not supported.
15	Generic Short Packet Code 8	R/W	When each bit is set to 1 and GSCT.GFIF = 1, data is output to the generic short packet FIFO.
14	Generic Short Packet Code 7	R/W	
13	Generic Short Packet Code 6	R/W	
12	Generic Short Packet Code 5	R/W	
11	Generic Short Packet Code 4	R/W	
10	Generic Short Packet Code 3	R/W	
9	Generic Short Packet Code 2	R/W	
8	Generic Short Packet Code 1	R/W	
7:5	Reserved	R	Fixed to 0.
4	End of Transmission Code	R/W	When this bit is set to 1b, EOTP packets are detected. Set to the same value as EMCT.EOTPEN.
3:0	Reserved	R	Fixed to 1.

67.3.11 DTEH : Receive Data Type Enable High Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x064

Bit position: 31

0

Bit field:

DTEN[63:32]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	DTEN[63:32]	Data Type Enable (DT = 0x20 to 0x3F) Receive data of a packet whose data type is one of the following. For details, see Table 67.3 . 0: Does not receive data 1: Receives data	R/W

Note: S-TYPE-3, P-TYPE-3

Table 67.3 Received data type of a packet for DTEN[63:32]

Bit	Data type	Attribute	Comment
31:11	Reserved	R/W	Changing the reserved bit from the initial value (0b) is prohibited.
10	RAW8	R/W	When this bit is set to 1b, data is output to the Video Pixel interface.
9:5	Reserved	R/W	Changing the reserved bit from the initial value (0b) is prohibited.
4	RGB888	R/W	When this bit is set to 1b, data is output to the Video Pixel interface.
3:0	Reserved	R/W	Changing the reserved bit from the initial value (0b) is prohibited.

67.3.12 RXST : Receive Status Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x070

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RACT DET	RACT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	FRM1 5	FRM1 4	FRM1 3	FRM1 2	FRM1 1	FRM1 0	FRM9	FRM8	FRM7	FRM6	FRM5	FRM4	FRM3	FRM2	FRM1	FRM0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FRM0	Frame of virtual channel 0 active Indicates that frames on virtual channel 0 are being processed.	R
1	FRM1	Frame of virtual channel 1 active Indicates that frames on virtual channel 1 are being processed.	R
2	FRM2	Frame of virtual channel 2 active Indicates that frames on virtual channel 2 are being processed.	R
3	FRM3	Frame of virtual channel 3 active Indicates that frames on virtual channel 3 are being processed.	R
4	FRM4	Frame of virtual channel 4 active Indicates that frames on virtual channel 4 are being processed.	R
5	FRM5	Frame of virtual channel 5 active Indicates that frames on virtual channel 5 are being processed.	R
6	FRM6	Frame of virtual channel 6 active Indicates that frames on virtual channel 6 are being processed.	R
7	FRM7	Frame of virtual channel 7 active Indicates that frames on virtual channel 7 are being processed.	R
8	FRM8	Frame of virtual channel 8 active Indicates that frames on virtual channel 8 are being processed.	R

Bit	Symbol	Function	R/W
9	FRM9	FRaMe of virtual channel 9 active Indicates that frames on virtual channel = 9 are being processed.	R
10	FRM10	Frame of virtual channel 10 active Indicates that frames on virtual channel 10 are being processed.	R
11	FRM11	Frame of virtual channel 11 active Indicates that frames on virtual channel 11 are being processed.	R
12	FRM12	Frame of virtual channel 12 active Indicates that frames on virtual channel 12 are being processed.	R
13	FRM13	Frame of virtual channel 13 active Indicates that frames on virtual channel 13 are being processed.	R
14	FRM14	Frame of virtual channel 14 active Indicates that frames on virtual channel 14 are being processed.	R
15	FRM15	Frame of virtual channel 15 active Indicates that frames on virtual channel 15 are being processed.	R
16	RACT	Rx (Reception) active status Indicates that this module is receiving packets.	R
17	RACTDET	RX (Reception) Active Detect When this bit is set to 1, RXST.RACT = 1 has been detected. Write 1 to RXSC.RACTDETC to clear this bit.	R
31:18	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

67.3.13 RXSC : Receive Status Clear Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x074

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RACT DETC	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
16:0	—	The write value should be 0.	W
17	RACTDETC	RX (Reception) Active Detect Status Clear Write 1 to this bit to clear RXST.RACTDET. This bit cannot be modified to 0.	W
31:18	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

67.3.14 RXIE : Receive Interrupt Enable Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x078

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RACTDETE	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
16:0	—	These bits are read as 0. The write value should be 0.	R/W
17	RACTDETE	RX (Reception) Active Detect Interrupt Enable Enables interrupts generated by RXST.RACTDET. 0: Does not assert csi2_int_rx when RXST.RACTDET = 1 1: Asserts csi2_int_rx when RXST.RACTDET = 1	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.3.15 DLST(N) : Data Lane (N) Status Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x080 + 0x010 × N (N = 0, 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	ULP	—	—	—	—	—	—	RUL	EUL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	EES	ECT	ESS	ESH
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ESH	ErrSotHs detect on data lane (N) status When this bit is set to 1, an SoT error (ErrSotHS) has occurred on data lane (N). Write 1 to DLSC(N).ESHC to clear this bit.	R
1	ESS	ErrSotSynchs detect on data lane (N) status When this bit is set to 1, an SoT synchronization error (ErrSotSynchHS) has occurred on data lane (N). Write 1 to DLSC(N).ESSC to clear this bit.	R
2	ECT	ErrConTrol detect on data lane (N) status When this bit is set to 1, a control error (ErrControl) has occurred on data lane (N). Write 1 to DLSC(N).ECTC to clear this bit.	R
3	EES	ErrESc detect on data lane (N) status When this bit is set to 1, an escape mode entry error (ErrESC) has occurred on data lane (N). Write 1 to DLSC(N).EESC to clear this bit.	R
15:4	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
16	EUL	Exit from ULps detect on data lane (N) status When this bit is set to 1, data lane (N) has exited from the ULPS receive mode (RxULpsEsc). Write 1 to DLSC(N).EULC to clear this bit.	R
17	RUL	Entry to ULPS detect on data lane (N) status When this bit is set to 1, data lane (N) has transitioned to the ULPS receive mode (RxULpsEsc). Write 1 to DLSC(N).RULC to clear this bit.	R
23:18	—	These bits are read as 0.	R
24	ULP	RxULPsc of data lane (N) status Indicates the RxULpsEsc status of data lane (N).	R
31:25	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

(N) indicates a data lane number.

(N) = 0, 1

67.3.16 DLSC(N) : Data Lane (N) Status Clear Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x084 + 0x010 × N (N = 0, 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RULC	EULC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	EESC	ECTC	ESSC	ESHC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ESHC	ErrSotHs detect on data lane (N) status Clear Write 1 to this bit to clear DLST(N).ESH. This bit cannot be modified to 0.	W
1	ESSC	ErrSotSynchs detect on data lane (N) status Clear Write 1 to this bit to clear DLST(N).ESS. This bit cannot be modified to 0.	W
2	ECTC	ErrConTrol detect on data lane (N) status Clear Write 1 to this bit to clear DLST(N).ECT. This bit cannot be modified to 0.	W
3	EESC	ErrESc detect on data lane (N) status Clear Write 1 to this bit to clear DLST(N).EES. This bit cannot be modified to 0.	W
15:4	—	The write value should be 0.	W
16	EULC	Exit from ULps detect on data lane (N) status Clear Write 1 to this bit to clear DLST(N).EUL. This bit cannot be modified to 0.	W
17	RULC	Entry to ULps detect on data lane (N) status Clear Write 1 to this bit to clear DLST(N).RUL. This bit cannot be modified to 0.	W
31:18	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

(N) indicates a data lane number.

(N) = 0, 1

67.3.17 DLIE(N) : Data Lane (N) Interrupt Enable Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x088 + 0x010 × N (N = 0, 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RULE	EULE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	EESE	ECTE	ESSE	ESHE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ESHE	ErrSotHs detect on data lane (N) interrupt Enable Enables interrupts generated by DLST(N).ESH. 0: Does not assert csi2_int_dl when DLST(N).ESH = 1 1: Asserts csi2_int_dl when DLST(N).ESH = 1	R/W
1	ESSE	ErrSotSynchs detect on data lane (N) interrupt Enable Enables interrupts generated by DLST(N).ESS. 0: Does not assert csi2_int_dl when DLST(N).ESS = 1 1: Asserts csi2_int_dl when DLST(N).ESS = 1	R/W
2	ECTE	ErrConTrol detect on data lane (N) interrupt Enable Enables interrupts generated by DLST(N).ECT. 0: Does not assert csi2_int_dl when DLST(N).ECT = 1 1: Asserts csi2_int_dl when DLST(N).ECT = 1	R/W
3	EESE	ErrESc detect on data lane (N) interrupt Enable Enables interrupts generated by DLST(N).EES. 0: Does not assert csi2_int_dl when DLST(N).EES = 1 1: Asserts csi2_int_dl when DLST(N).EES = 1	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W
16	EULE	Exit to ULps detect on data lane (N) interrupt Enable Enables interrupts generated by DLST(N).EUL. 0: Does not assert csi2_int_dl when DLST(N).EUL = 1 1: Asserts csi2_int_dl when DLST(N).EUL = 1	R/W
17	RULE	Entry to ULps detect on data lane (N) interrupt Enable Enables interrupts generated by DLST(N).RUL. 0: Does not assert csi2_int_dl when DLST(N).RUL = 1 1: Asserts csi2_int_dl when DLST(N).RUL = 1	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

(N) indicates a data lane number.

(N) = 0, 1

67.3.18 VCST(M) : Virtual Channel (M) Status Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x100 + 0x010 × M (M = 0 to 15)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	ETR	LER	LSR	FER	FSR	—	—	—	—	—	—	—	OVF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	FRD	FRS	—	ECN	ECC	WCE	IDE	CRC	ECD	MLF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MLF	MaLFormed packet with virtual channel (M) detect status When this bit is set to 1, a packet of less than 4 bytes has been received. Because the virtual channel of the packet with an error cannot be determined, the MLF field is set to 1 for all virtual channels. To clear the MLF field for all virtual channels, write 1 to VCSC(M).AMLFC for any virtual channel. To clear the MLF field of virtual channel M, write 1 to VCSC(M).MLFC.	R
1	ECD	ECc 2-bit (Double) error packet with virtual channel (M) Detect status When this bit is set to 1, a 2-bit error was found in the ECC check on the received packets. Because the virtual channel of the packet with an error cannot be determined, the ECD field is set to 1 for all virtual channels. To clear the ECD field for all virtual channels, write 1 to VCSC(M).AECD for any virtual channel. To clear the ECD field of virtual channel M, write 1 to VCSC(M).ECDC.	R
2	CRC	CRC error packet with virtual channel (M) detect status When this bit is set to 1, a CRC error was detected in the received packet. Write 1 to VCSC(M).CRCC to clear this bit.	R
3	IDE	ErrID packet with virtual channel (M) detect status When this bit is set to 1, an ErrID was detected and the packet was discarded. This bit is set to 1 if any of the following conditions occurs: - A packet which is defined as "Reserved or Unavailable" in chapter 9, Low Level Protocol in the CSI-2 Specification has been received. - A packet whose data type is 0x05 to 0x07, 0x3F has been received. - A packet whose data type is 0x39 to 0x3E has been received when MCT0.RVMD = 0. - A packet whose data type is 0x04 has been received when EMCT.EOTPEN = 0. - A generic short packet (Data Type = 0x08 to 0x0F) for which the corresponding bit in DTEL.DTEN is 0 has been received. - A long packet (Data Type = 0x10 to 0x3E, also includes 0x38 (USL Command)) for which the corresponding bit in DTEL(H).DTEN is 0 has been received. Write 1 to VCSC(M).IDEC to clear this bit. Note that this bit may be set to 1 if more Spacers than the count specified in EPCT.SSP/SLP are received when using EPD Option 2 fixed length Spacer selection (EPCT.EPDEN = 1 & EPDOP = 1 & EMCT.VLSIEN = 00b).	R
4	WCE	Word Count Error packet with virtual channel (M) detect status When this bit is set to 1, it was detected that the data payload length of the packet is shorter than the value indicated by WC. Write 1 to VCSC(M).WCEC to clear this bit.	R
5	ECC	ECc 1-bit error (Corrected) packet with virtual channel (M) detect status When this bit is set to 1, a 1-bit error was found in the ECC check on the received packets and the error was corrected. Write 1 to VCSC(M).ECCC to clear this bit.	R
6	ECN	ECc No-error packet with virtual channel (M) detect status When this bit is set to 1, no error was found in the ECC check on the received packets. Write 1 to VCSC(M).ECNC to clear this bit.	R
7	—	This bit is read as 0.	R

Bit	Symbol	Function	R/W
8	FRS	errFrameSync of virtual channel (M) detect status When this bit is set to 1, ErrFrameSync has been detected. This bit is set to 1 if any of the following conditions occurs: • After an FS is received, another FS has been received on the same virtual channel without receiving an FE. • Although an FE is received, an FS has not been received on the same virtual channel. Write 1 to VCSC(M).FRSC to clear this bit.	R
9	FRD	errFrameData of virtual channel (M) detect status When this bit is set to 1, ErrFrameData has been detected. This bit is set to 1 if any of the following errors occurs between FS and FE: • ErrCRC • ErrWC • ECC 2-bit error (when MCT0.EDMD = 1) • Reception of a packet of less than four bytes (when MCT0.EDMD = 1) Note that this bit is set to 1 if Virtual Channel = M is between FS and FE because the virtual channel of a packet with an ECC 2-bit error or the virtual channel which received a packet of less than four bytes cannot be determined. Write 1 to VCSC(M).FRDC to clear this bit.	R
15:10	—	These bits are read as 0.	R
16	OVF	generic short packet with virtual channel (M) discard by fifo OVerFlow status When this bit is set to 1, the generic short packets on virtual channel M has been discarded due to an overflow of the generic short packet FIFO. Write 1 to VCSC(M).OVFC to clear this bit.	R
23:17	—	These bits are read as 0.	R
24	FSR	Frame Start packet with virtual channel (M) Reception status When this bit is set to 1, a packet whose data type is "Frame Start Code" has been received. Write 1 to VCSC(M).FSRC to clear this bit.	R
25	FER	Frame End packet with virtual channel (M) Reception status When this bit is set to 1, a packet whose data type is "Frame End Code" has been received. Write 1 to VCSC(M).FERC to clear this bit.	R
26	LSR	Line Start packet with virtual channel (M) Reception status When this bit is set to 1, a packet whose data type is "Line Start Code" has been received. Write 1 to VCSC(M).LSRC to clear this bit.	R
27	LER	Line End packet with virtual channel (M) Reception status When this bit is set to 1, a packet whose data type is "Line End Code" has been received. Write 1 to VCSC(M).LERC to clear this bit.	R
28	ETR	EoTp packet with virtual channel (M) Reception status When this bit is set to 1, a packet whose Data Type = End-of-Transmission Code (EoTp) has been received. This bit is set only when EMCT.EOTPEN = 1. Since the EoTp Virtual Channel is specified as 0 in the CSI-2 standard, ETR1 to ETR15 are not usually set. Write 1 to VCSC (M) .ETRC to clear this bit.	R
31:29	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

(M) indicates a virtual channel number.

(M) 0, 1, 2, ..., 14, 15

67.3.19 VCSC(M) : Virtual Channel (M) Status Clear Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x104 + 0x010 × M (M = 0 to 15)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	ETRC	LERC	LSRC	FERC	FSRC	—	—	—	—	—	—	—	OVFC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AECD C	AMLF C	—	—	—	—	FRDC	FRSC	—	ECNC	ECCC	WCEC	IDEC	CRCC	ECDC	MLFC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MLFC	MaLFormed packet with virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).MLF. This bit cannot be modified to 0.	W
1	ECDC	ECc 2-bit (Double) error packet with virtual channel (M) Detect status Clear Write 1 to this bit to clear VCST(M).ECD. This bit cannot be modified to 0.	W
2	CRCC	CRC error packet with virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).CRC. This bit cannot be modified to 0.	W
3	IDEC	ErrID packet with virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).IDE. This bit cannot be modified to 0.	W
4	WCEC	Word Count Error packet with virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).WCE. This bit cannot be modified to 0.	W
5	ECCC	ECc 1-bit error (Corrected) packet with virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).ECC. This bit cannot be modified to 0.	W
6	ECNC	ECc No-error packet with virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).ECN. This bit cannot be modified to 0.	W
7	—	The write value should be 0.	W
8	FRSC	errFrameSync of virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).FRS. This bit cannot be modified to 0.	W
9	FRDC	errFrameData of virtual channel (M) detect status Clear Write 1 to this bit to clear VCST(M).FRD. This bit cannot be modified to 0.	W
13:10	—	The write value should be 0.	W
14	AMLFC	MaLFormed packet with any virtual channels detect status Clear Write 1 to this bit to clear VCST(M).MLF for all virtual channels. This bit cannot be modified to 0.	W
15	AECD	ECc 2-bit (Double) error packet with any virtual channels Detect status Clear Write 1 to this bit to clear VCST(M).ECD for all virtual channels. This bit cannot be modified to 0.	W
16	OVFC	Generic short packet with virtual channel (M) discard by FIFO Overflow Status Clear Write 1 to this bit to clear VCST(M).OVF. This bit cannot be modified to 0.	W
23:17	—	The write value should be 0.	W

Bit	Symbol	Function	R/W
24	FSRC	Frame Start packet with virtual channel (M) Reception status Clear Write 1 to this bit to clear VCST(M).FSR. This bit cannot be modified to 0.	W
25	FERC	Frame End packet with virtual channel (M) Reception status Clear Write 1 to this bit to clear VCST(M).FER. This bit cannot be modified to 0.	W
26	LSRC	Line Start packet with virtual channel (M) Reception status Clear Write 1 to this bit to clear VCST(M).LSR. This bit cannot be modified to 0.	W
27	LERC	Line End packet with virtual channel (M) Reception status Clear Write 1 to this bit to clear VCST(M).LER. This bit cannot be modified to 0.	W
28	ETRC	EoTp packet with virtual channel (M) Reception status Clear Write 1 to this bit to clear VCST(M).ETR. This bit cannot be modified to 0.	W
31:29	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

(M) indicates a virtual channel number.

(M) = 0, 1, 2, ..., 14, 15

67.3.20 VCIE(M) : Virtual Channel (M) Interrupt Enable Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x108 + 0x010 × M (M = 0 to 15)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	ETRE	LERE	LSRE	FERE	FSRE	—	—	—	—	—	—	—	OVFE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	FRDE	FRSE	—	ECNE	ECCE	WCEE	IDEE	CRCE	ECDE	MLFE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MLFE	MaLFormed packet with virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).MLF. 0: Does not assert csi2_int_vc when VCST(M).MLF = 1 1: Asserts csi2_int_vc when VCST(M).MLF = 1	R/W
1	ECDE	ECc 2-bit (Double) error packet with virtual channel (M) Detect interrupt Enable Enables interrupts generated by VCST(M).ECD. 0: Does not assert csi2_int_vc when VCST(M).ECD = 1 1: Asserts csi2_int_vc when VCST(M).ECD = 1	R/W
2	CRCE	CRC error packet with virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).CRC. 0: Does not assert csi2_int_vc when VCST(M).CRC = 1 1: Asserts csi2_int_vc when VCST(M).CRC = 1	R/W
3	IDEE	ErrID packet with virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).IDE. 0: Does not assert csi2_int_vc when VCST(M).IDE = 1 1: Asserts csi2_int_vc when VCST(M).IDE = 1	R/W

Bit	Symbol	Function	R/W
4	WCEE	Word Count Error packet with virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).WCE. 0: Does not assert csi2_int_vc when VCST(M).WCE = 1 1: Asserts csi2_int_vc when VCST(M).WCE = 1	R/W
5	ECCE	ECc 1-bit error (Corrected) packet with virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).ECC. 0: Does not assert csi2_int_vc when VCST(M).ECC = 1 1: Asserts csi2_int_vc when VCST(M).ECC = 1	R/W
6	ECNE	ECc No-error packet with virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).ECN. 0: Does not assert csi2_int_vc when VCST(M).ECN = 1 1: Asserts csi2_int_vc when VCST(M).ECN = 1	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	FRSE	errFrameSync of virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).FRS. 0: Does not assert csi2_int_vc when VCST(M).FRS = 1 1: Asserts csi2_int_vc when VCST(M).FRS = 1	R/W
9	FRDE	errFrameData of virtual channel (M) detect interrupt Enable Enables interrupts generated by VCST(M).FRD. 0: Does not assert csi2_int_vc when VCST(M).FRD = 1 1: Asserts csi2_int_vc when VCST(M).FRD = 1	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
16	OVFE	Generic short packet with virtual channel (M) discarded by FIFO Overflow interrupt Enable Enables interrupts generated by VCST(M).OVF. 0: Does not assert csi2_int_vc when VCST(M).OVF = 1 1: Asserts csi2_int_vc when VCST(M).OVF = 1	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	FSRE	Frame Start packet with virtual channel (M) Reception interrupt Enable Enables interrupts generated by VCST(M).FSR. 0: Does not assert csi2_int_vc when VCST(M).FSR = 1 1: Asserts csi2_int_vc when VCST(M).FSR = 1	R/W
25	FERE	Frame End packet with virtual channel (M) Reception interrupt Enable Enables interrupts generated by VCST(M).FER. 0: Does not assert csi2_int_vc when VCST(M).FER = 1 1: Asserts csi2_int_vc when VCST(M).FER = 1	R/W
26	LSRE	Line Start packet with virtual channel (M) Reception interrupt Enable Enables interrupts generated by VCST(M).LSR. 0: Does not assert csi2_int_vc when VCST(M).LSR = 1 1: Asserts csi2_int_vc when VCST(M).LSR = 1	R/W
27	LERE	Line End packet with virtual channel (M) Reception interrupt Enable Enables interrupts generated by VCST(M).LER. 0: Does not assert csi2_int_vc when VCST(M).LER = 1 1: Asserts csi2_int_vc when VCST(M).LER = 1	R/W
28	ETRE	EoTp packet with virtual channel (M) Reception interrupt Enable Enables interrupts generated by VCST(M).ETR. 0: Does not assert csi2_int_vc when VCST(M).ETR = 1 1: Asserts csi2_int_vc when VCST(M).ETR = 1	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

(M) indicates a virtual channel number.

(M) = 0, 1, 2, ..., 14, 15

67.3.21 PMST : Power Management Status Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x200

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	DLUL	—	—	—	—	—	—	—	DLSS	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CLUL	CLSS	—	—	—	—	—	—	CUN	CUX	DUN	DUX	CSN	CSX	DSN	DSX
Value after reset:	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DSX	eXit from Stop state detect on all valid Data lanes status When this bit is set to 1, the mode of all valid data lanes has changed from the stop state to a different state. Write 1 to PMSC.DSXC to clear this bit.	R
1	DSN	eNtry to Stop state detect on all valid Data lanes status When this bit is set to 1, all valid data lanes have transitioned to the stop state. Write 1 to PMSC.DSNC to clear this bit.	R
2	CSX	eXit from Stop state detect on Clock lane status When this bit is set to 1, the state of the clock lane has changed from the stop state to a state different from the stop state. Write 1 to PMSC.CSXC to clear this bit.	R
3	CSN	eNtry to Stop state detect on Clock lane status When this bit is set to 1, the clock lane has transitioned to the stop state. Write 1 to PMSC.CSNC to clear this bit.	R
4	DUX	eXit from Ulps detect on all valid Data lanes status When this bit is set to 1, the mode of all valid data lanes has changed from ULPS mode to a different state. Write 1 to PMSC.DUXC to clear this bit.	R
5	DUN	eNtry to Ulps detect on all valid Data lanes status When this bit is set to 1, all valid data lanes have transitioned to ULPS mode. Write 1 to PMSC.DUNC to clear this bit.	R
6	CUX	eXit from Ulps detect on Clock lane status When this bit is set to 1, the mode of the clock lane has changed from ULPS mode (inversion of RxUlpsClkNot) to non-ULPS mode. Write 1 to PMSC.CUXC to clear this bit.	R
7	CUN	eNtry to Ulps detect on Clock lane status When this bit is set to 1, the clock lane has transitioned to ULPS mode (inversion of RxUlpsClkNot). Write 1 to PMSC.CUNC to clear this bit.	R
13:8	—	These bits are read as 0.	R
14	CLSS	Stop State of Clock Lane status Indicates the stop state of the PPI clock lane. 0: Not in the stop state 1: In the stop state	R
15	CLUL	rxULpsclknot (inverted) of Clock Lane status Indicates the inverted state of RxUlpsClkNot of the PPI clock lane. 0: Not in ULPS 1: In ULPS	R
17:16	DLSS	Stop State of Data Lanes status Indicate the stop state of a data lane. 0: Stop state of lane 0 1: Stop state of lane 1	R

Bit	Symbol	Function	R/W
23:18	—	These bits are read as 0.	R
25:24	DLUL	RxULpsesc of Data Lanes status Indicate the RxULpsEsc status of a data lane. 0: RxULpsEsc of lane 0 1: RxULpsEsc of lane 1	R
31:26	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

67.3.22 PMSC : Power Management Status Clear Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x204

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	CUNC	CUXC	DUNC	DUXC	CSNC	CSXC	DSNC	DSXC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DSXC	eXit from Stop state detect on all valid Data lanes status Clear Write 1 to this bit to clear PMST.DSX. This bit cannot be modified to 0.	W
1	DSNC	eNtry to Stop state detect on all valid Data lanes status Clear Write 1 to this bit to clear PMST.DSN. This bit cannot be modified to 0.	W
2	CSXC	eXit from Stop state detect on Clock lane status Clear Write 1 to this bit to clear PMST.CSX. This bit cannot be modified to 0.	W
3	CSNC	eNtry to Stop state detect on Clock lane status Clear Write 1 to this bit to clear PMST.CSN. This bit cannot be modified to 0.	W
4	DUXC	eXit from Ulps detect on all valid Data lanes status Clear Write 1 to this bit to clear PMST.DUX. This bit cannot be modified to 0.	W
5	DUNC	eNtry to Ulps detect on all valid Data lanes status Clear Write 1 to this bit to clear PMST.DUN. This bit cannot be modified to 0.	W
6	CUXC	eXit from Ulps detect on Clock lane status Clear Write 1 to this bit to clear PMST.CUX. This bit cannot be modified to 0.	W
7	CUNC	eNtry to Ulps detect on Clock lane status Clear Write 1 to this bit to clear PMST.CUN. This bit cannot be modified to 0.	W
31:8	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

67.3.23 PMIE : Power Management Interrupt Enable Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x208

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	CUNE	CUXE	DUNE	DUXE	CSNE	CSXE	DSNE	DSXE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DSXE	eXit from Stop state detect on all valid Data lanes interrupt Enable Enables interrupts generated by PMST.DSX. 0: Does not assert csi2_int_pm when PMST.DSX = 1. 1: Asserts csi2_int_pm when PMST.DSX = 1.	R/W
1	DSNE	eNtry to Stop state detect on all valid Data lanes interrupt Enable Enables interrupts generated by PMST.DSN. 0: Does not assert csi2_int_pm when PMST.DSN = 1. 1: Asserts csi2_int_pm when PMST.DSN = 1.	R/W
2	CSXE	eXit from Stop state detect on Clock lane interrupt Enable Enables interrupts generated by PMST.CSX. 0: Does not assert csi2_int_pm when PMST.CSX = 1. 1: Asserts csi2_int_pm when PMST.CSX = 1.	R/W
3	CSNE	eNtry to Stop state detect on Clock lane interrupt Enable Enables interrupts generated by PMST.CSN. 0: Does not assert csi2_int_pm when PMST.CSN = 1. 1: Asserts csi2_int_pm when PMST.CSN = 1.	R/W
4	DUXE	eXit from Ulps detect on all valid Data lanes interrupt Enable Enables interrupts generated by PMST.DUX. 0: Does not assert csi2_int_pm when PMST.DUX = 1. 1: Asserts csi2_int_pm when PMST.DUX = 1.	R/W
5	DUNE	eNtry to Ulps detect on all valid Data lanes interrupt Enable Enables interrupts generated by PMST.DUN. 0: Does not assert csi2_int_pm when PMST.DUN = 1. 1: Asserts csi2_int_pm when PMST.DUN = 1.	R/W
6	CUXE	eXit from Ulps detect on Clock lane interrupt Enable Enables interrupts generated by PMST.CUX. 0: Does not assert csi2_int_pm when PMST.CUX = 1. 1: Asserts csi2_int_pm when PMST.CUX = 1.	R/W
7	CUNE	eNtry to Ulps detect on Clock lane interrupt Enable Enables interrupts generated by PMST.CUN. 0: Does not assert csi2_int_pm when PMST.CUN = 1. 1: Asserts csi2_int_pm when PMST.CUN = 1.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.3.24 GSCT : Generic Short Packet Control Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x280

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GFIF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	SHTH						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	SHTH	Stored generic short packet THreshold Asserts GSST.GTH while the number of packets stored in the generic short packet FIFO equals or exceeds the value of this field + 1. Set this field to 15 or less when using GSST.GTH. Set the default value when not using GSST.GTH.	R/W
15:7	—	These bits are read as 0. The write value should be 0.	R/W
16	GFIF	Generic short packet store in FIFO When this bit is set to 1, the received generic short packet is stored in the generic short packet FIFO (GS FIFO).	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.3.25 GSST : Generic Short Packet Status Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x284

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STRD S	GCD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PNUM								—	—	—	GOV	—	—	GTH	GNE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GNE	Generic short packet FIFO Not Empty When this bit is set to 1, the generic short packet FIFO is not empty.	R
1	GTH	More than THreshold Generic short packets existed in FIFO When this bit is set to 1, the number of packets stored in the generic short packet FIFO equals or exceeds GSCT.SHTH + 1.	R
3:2	—	These bits are read as 0.	R
4	GOV	Generic short packet fifo OVerflow status When this bit is set to 1, the generic short packet FIFO has overflowed. Write 1 to GSSC.GOV to clear this bit.	R
7:5	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
15:8	PNUM	NUMBER of stored generic short Packets in FIFO Indicate the number of packets stored in the generic short packet FIFO.	R
16	GCD	Generic short packet FIFO Clear status When this bit is set to 1, the GS FIFO is cleared by GSCT.GFCLR.	R
17	STRDS	Generic short packet SToRe DiSable Indicates whether the received generic short packets can be stored in the generic short packet FIFO. This field indicates an inverted value of the store_en signal described in section 67.4.1. Start/Stop of Reception Operation with a delay of internal latency. It may indicate the previous value depending on the signal change timing. 0: Can be stored in the generic short packet FIFO 1: Cannot be stored in the generic short packet FIFO	R
31:18	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

67.3.26 GSSC : Generic Short Packet Status Clear Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x288

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	GOVC	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	The write value should be 0.	W
4	GOVC	Generic short packet FIFO OVERflow status Clear Write 1 to this bit to clear GSST.GOV. This bit cannot be modified to 0.	W
31:5	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

67.3.27 GSIE : Generic Short Packet Interrupt Enable Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x28C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	GOVE	—	—	GTHE	GNEE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GNEE	Generic short packet FIFO Not Empty interrupt Enable Enables interrupts generated by GSST.GNE. 0: Does not assert <code>csi2_int_gst</code> when GSST.GNE = 1 1: Asserts <code>csi2_int_gst</code> when GSST.GNE = 1	R/W
1	GTHE	More than THreshold Generic short packets existed in FIFO interrupt Enable Enables interrupts generated by GSST.GTH. 0: Does not assert <code>csi2_int_gst</code> when GSST.GTH = 1 1: Asserts <code>csi2_int_gst</code> when GSST.GTH = 1	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	GOVE	Generic short packet FIFO OVERflow interrupt Enable Enables interrupts generated by GSST.GOV. 0: Does not assert <code>csi2_int_gst</code> when GSST.GOV = 1 1: Asserts <code>csi2_int_gst</code> when GSST.GOV = 1	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.3.28 GSHT : Generic Short Packet Register

Base address: MIPI_CSI0 = 0x4034_7000
MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x290

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	SPVC				—	—	DTYP					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SPDT															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	SPDT	Stored Packet DaTa Indicate 16-bit user-defined data.	R
21:16	DTYP	Stored packet Data TYPE Indicate the data type. 0x08 to 0x0F	R
23:22	—	These bits are read as 0.	R
27:24	SPVC	Stored Packet Virtual Channel Indicate the virtual channel.	R
31:28	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

67.3.29 GSIU : Generic Short Packet Information Update Register

Base address: MIPI_CSI0 = 0x4034_7000
 MIPI_CSI0_NS = 0x5034_7000

Offset address: 0x294

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GFEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	GFCLR	—	—	—	—	—	—	—	FINC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FINC	Generic short packet FIFO update (INCrement internal pointer) Write 1 to this bit to update the next packet with the data of the packet indicated in the GSHT register. This bit cannot be modified to 0. Writing 1 to this bit while GSST.PNUM = 0 is prohibited.	W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	GFCLR	Generic short packet FIFO CLear Clears the GS FIFO. Writing 1 to this bit requests the FIFO to be cleared. Writing 0 to this bit requests FIFO clearing to be canceled.	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	GFEN	Generic short packet FIFO ENable Write 1 to this bit to re-enable storing generic short packets in the GS FIFO which has overflowed and been disabled. This bit cannot be modified to 0.	W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

67.4 Operation

67.4.1 Start/Stop of Reception Operation

(1) Start of reception operation

Figure 67.1 shows the flow for starting the receive operation.

Initialize the registers with MCT3.RXEN = 0. Registers to be used with their initial values do not need to be set.

After completing the initialization of the registers, set MCT3.RXEN to 1 to start reception operation of this module.

The registers that are initially set in the flow shown in Figure 67.1 can only be changed when MCT3.RXEN = 0.

Changing them at any other timing is prohibited.

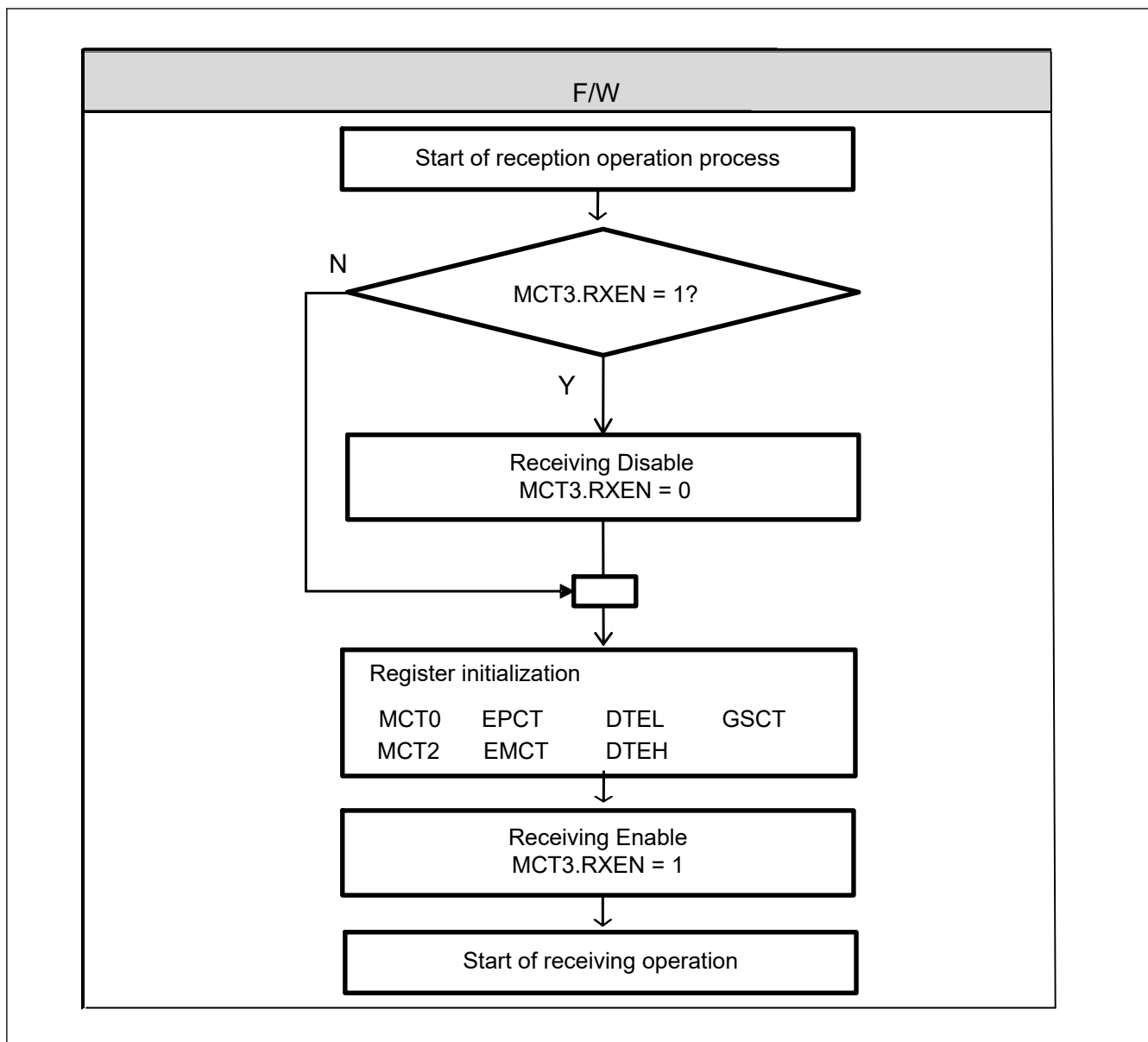


Figure 67.1 Start of receiving operation

(2) Receiving operation stopped

Figure 67.2 shows the flow of receiving operation stops.

Issue a stop request to the CSI-2 transmitter, such as by issuing an Enter Software Standby CCI command on the CCI.

The CSI-2 transmitter should stop within one frame time, but the CSI-2 receiver can only determine that the CSI-2 transmitter has stopped by not receiving packets for a certain period of time.

After waiting at least one frame, write 1 to RXSC.RACTDETC and clear RXST.RACTDET to 0. After waiting at least one frame time, the RXST is read.

Normally, RXST.RACTDET = 0 and RXST.FRM0-15 = 0, but the previously mentioned values may not be obtained in the following cases. Execute the restop process and error recovery process as necessary.

1. If for some reason the CSI-2 transmitter does not stop transmitting, RXST.RACTDET = 0 is not set. In this case, some abnormality is considered to have occurred on the CSI-2 transmitter side.
2. If the Frame End packet is lost due to ECC 2-bit error, for example, this module cannot recognize that the frame has ended, so RXST.FRM0-15 = 0 is not set.

Setting 0 to MCT3.RXEN disables reception of this module.

After disabling reception, perform a software reset as described in [section 67.4.3. Software Reset](#) to reset the signals output to the Video Pixel interface and complete the process of stopping reception operation.

After completion of the process to stop the receiving operation, the receiving operation can be started again by following the flow procedure shown in [Figure 67.1](#).

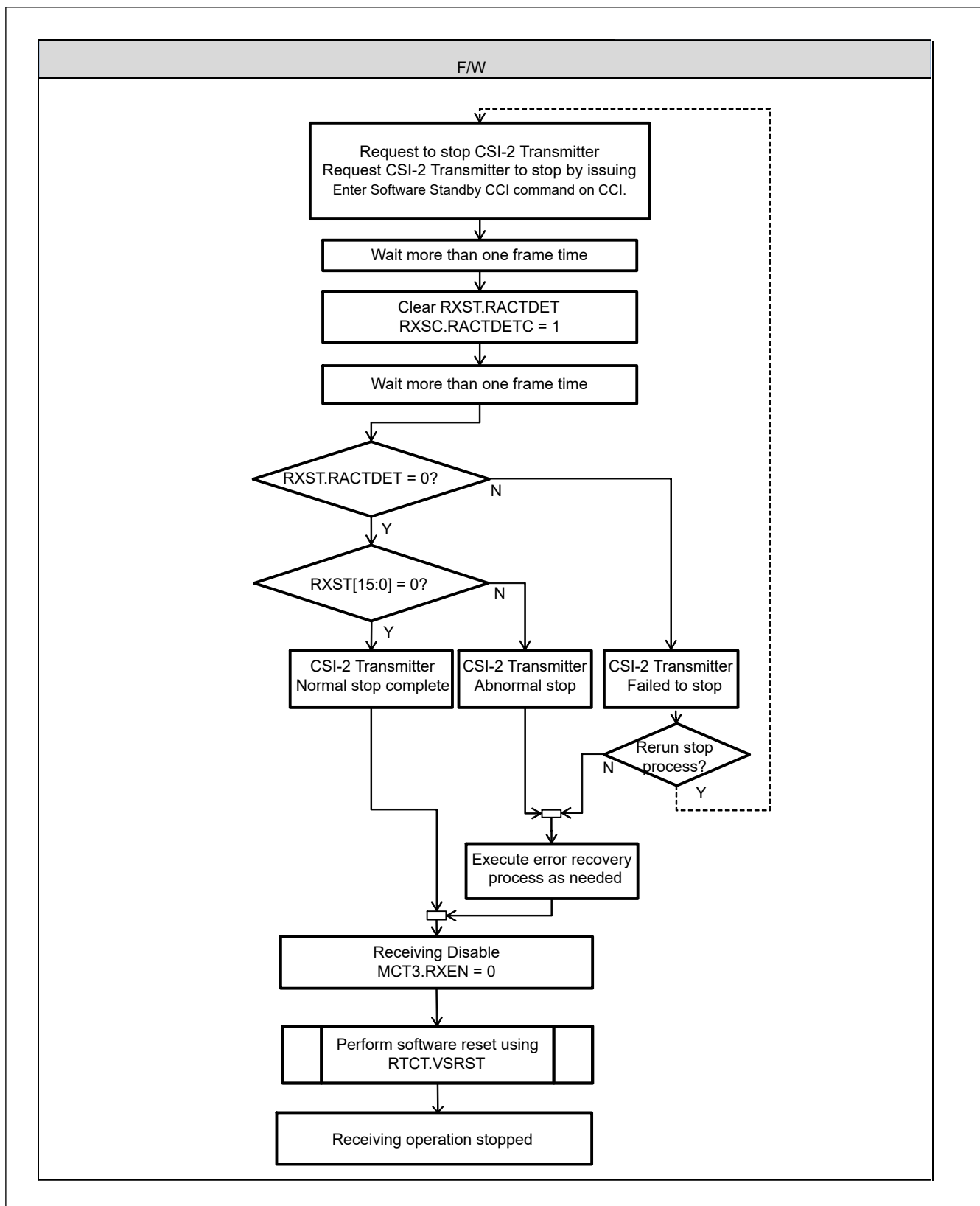


Figure 67.2 Receiving operation stopped

67.4.2 Data Reception

(1) Common Between Short Packets and Long Packets

Figure 67.3 shows the flow when receiving a packet whose data type is A and virtual channel is M.

This flow is common between short packets and long packets.

On receiving a packet, the module checks the length of the received packet.

If the length of the received packet is less than the packet header length (4 bytes), a virtual channel cannot be determined, resulting in setting VCST(M).MLF to 1 for all virtual channels. In addition, this packet is discarded because it is invalid.

When the length of the received packet is 4 bytes or more, ECC checking is performed. The check results are indicated in the VCST(M) register.

- 2-bit error: Because the virtual channel of the packet with an error cannot be determined, VCST(M).ECD is set to 1 for all virtual channels.
In addition, the packet is discarded because error correction cannot be performed.
- 1-bit error: VCST(M).ECC is set to 1 for the relevant virtual channel.
Error correction is performed because it is possible.
- No error: VCST(M).ECN is set to 1 for the relevant virtual channel.

If the ECC check results are 1-bit error or no error, the data type field of the packet header is analyzed.

If the Data Type is reserved according to Chapter 9 of the CSI-2 standard (however, if MCT0.RVMD = 1, Data Type = (0x39 to 0x3E) is regarded as a long packet), VCST(M).IDE set 1 to discard the packet.

For short packets, the process is described in (2) Short Packets. For long packets, the process is described in (3) Long Packets.

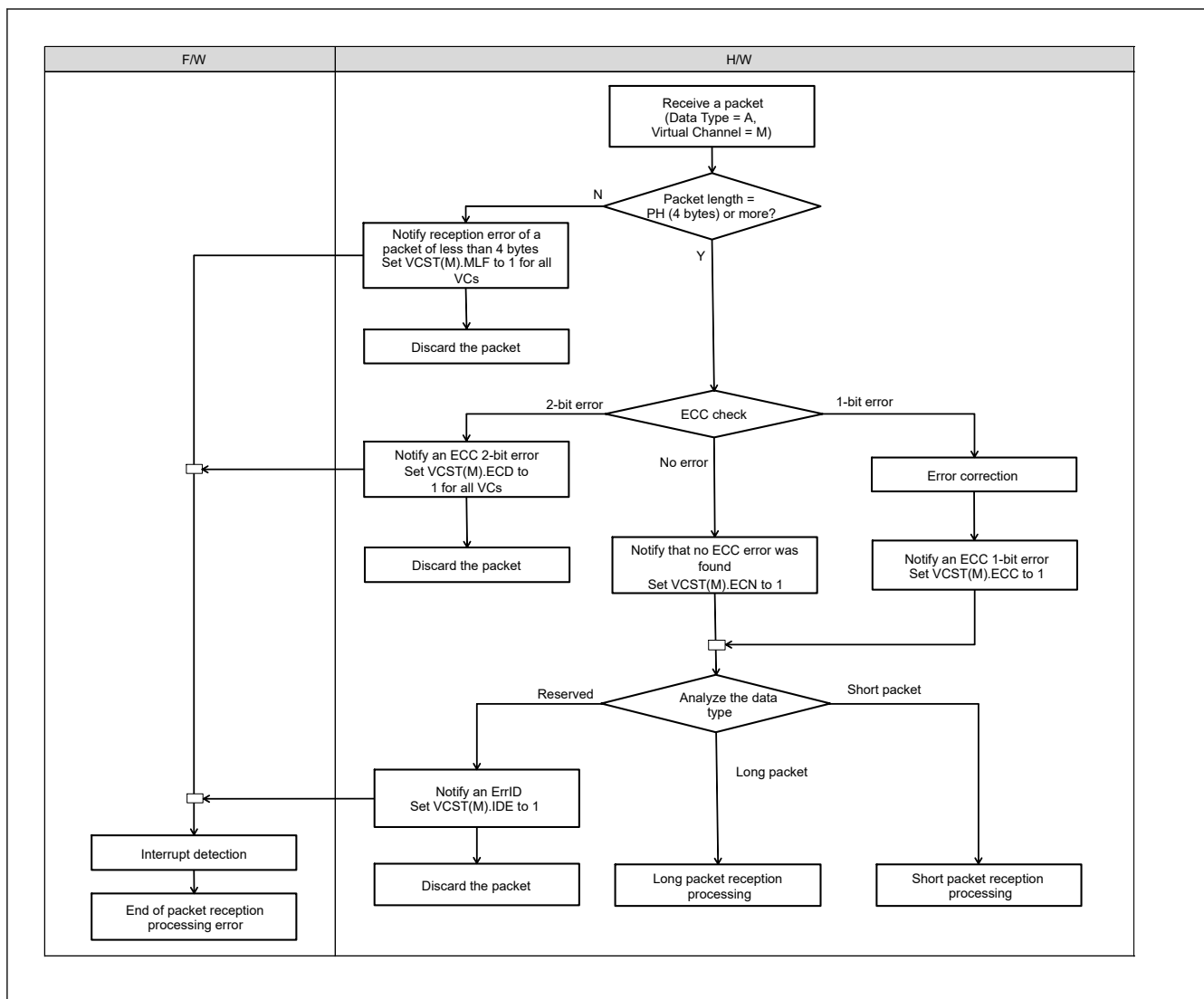


Figure 67.3 Packet reception processing

(2) Short Packets

(a) Frame Start Code

When a packet with Data Type = Frame Start Code and Virtual Channel = M is received, it is notified that it was received in the VCST(M).FSR field.

(b) Frame End Code

When a packet with Data Type = Frame End Code and Virtual Channel = M is received, it is notified that it was received in the VCST(M).FER field.

(c) Line Start Code

When a packet with Data Type = Line Start Code and Virtual Channel = M is received, it is notified that it was received in the VCST(M).LSR field.

(d) Line End Code

When a packet with Data Type = Line End Code and Virtual Channel = M is received, it is notified that it was received in the VCST(M).LER field.

(e) End of Transmission Code

When a packet with Data Type = End of Transmission Code and Virtual Channel = M is received, it is notified that it was received in the VCST(M).ETR field.

(f) Generic Short Packet Data Type Code

Figure 67.4 shows the flow when receiving a packet whose data type is generic short packet code 1 to 8 and virtual channel is M.

When the corresponding bit in DTEL.DTEN is set to 0, VCST(M).IDE is asserted and the packet is discarded.

When the corresponding bit in DTEL.DTEN is set to 1 and GSCT.GFIF is set to 1, the generic short packet is stored to the generic short packet FIFO. For details on the operation when using the generic short packet FIFO, see the flow in Figure 67.4.

The initial value of internal signal store_en which is described in the flow is 1. When store_en is set to 1, packets can be stored in the generic short packet FIFO. When it is set to 0, packets cannot be stored in the generic short packet FIFO.

- store_en is set to 0 when the generic short packet FIFO has overflowed.
- store_en is set to 1 when 1 is written to GSIU.GFEN.

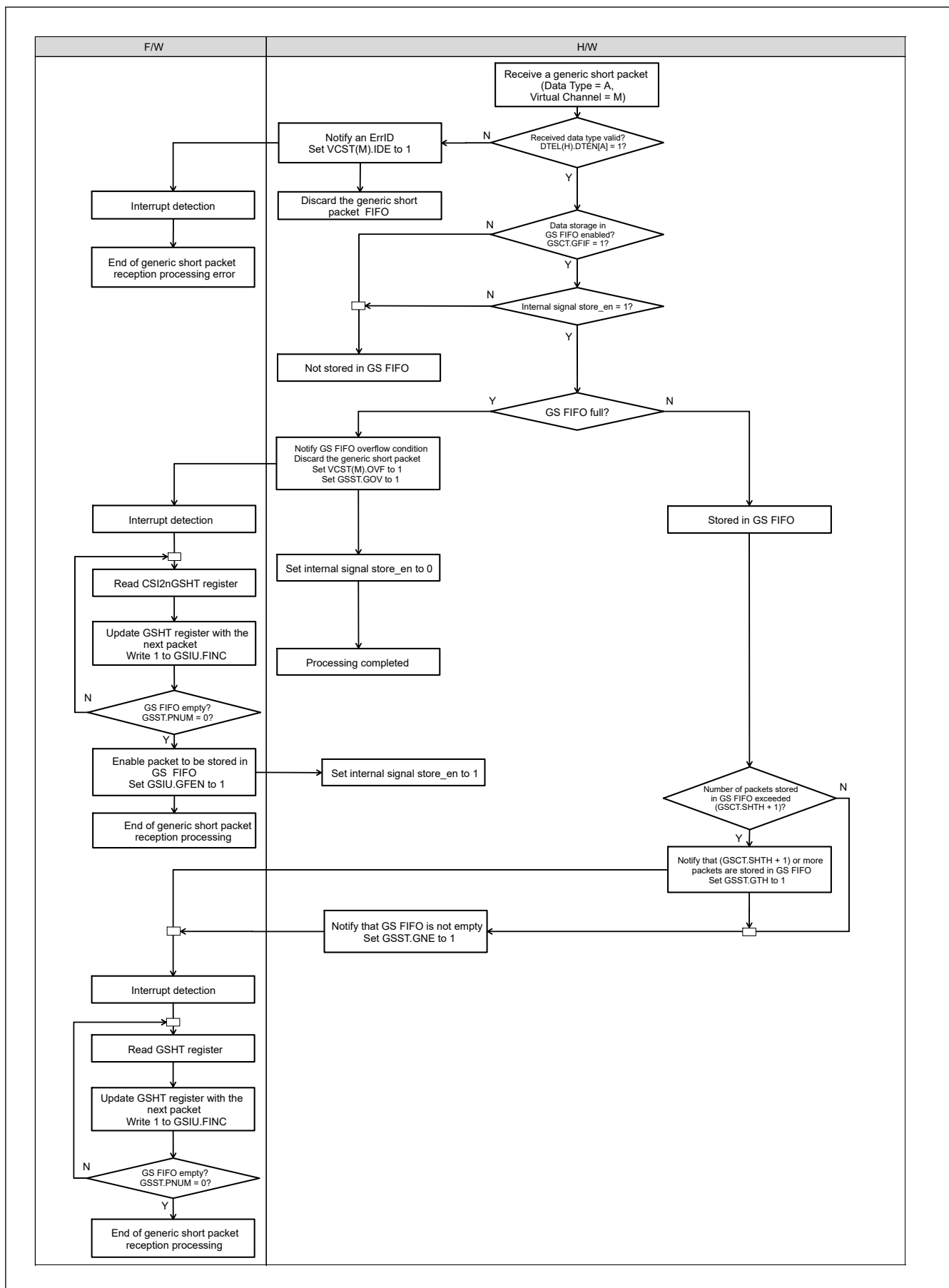


Figure 67.4 Generic short packet receive processing

(f).1 Clearing the Generic Short Packet FIFO

Figure 67.5 shows how the generic short packet FIFO is cleared.

Writing 1 to GSIU.GFCLR starts clearing the generic short packet FIFO.

After requesting to clear the generic short packet FIFO, check if GSST.GCD is set to 1 to make sure that the FIFO is cleared. Then, writing 0 to GSIU.GFCLR cancels the request for clearing the generic short packet FIFO. Check if GSST.GCD is set to 0 to make sure that the request for clearing the generic short packet FIFO is canceled.

When MCG.GSNM is set to 0d, it is prohibited to clear the generic short packet FIFO.

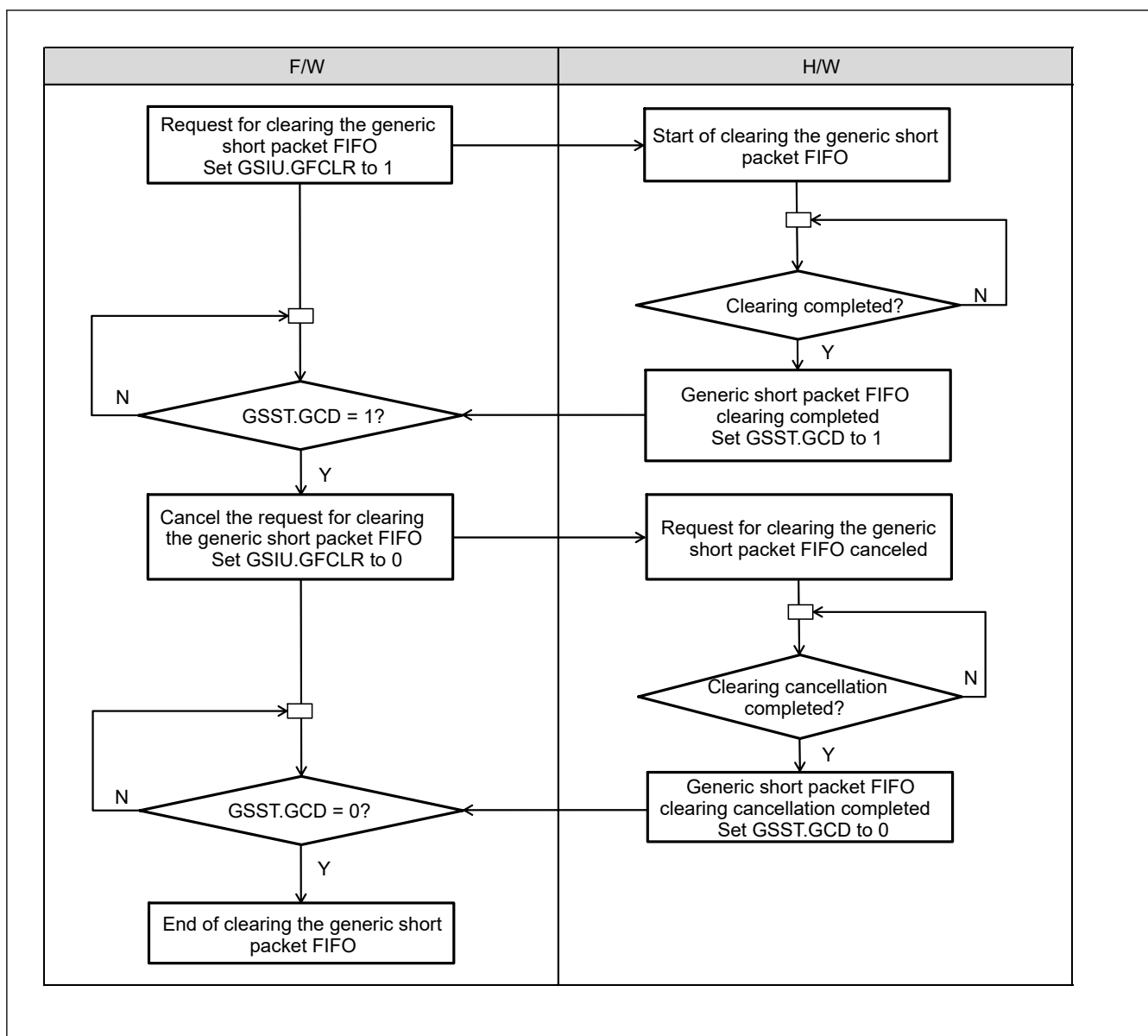


Figure 67.5 Clearing the generic short packet FIFO

(3) Long Packets

Figure 67.6 shows the flow when a long packet is received.

The following operations are performed according to the relevant DTEH.DTEN/DTEL.DTEN value.

0: Discards the received long packet and sets 1 to VCST(M).IDE.

1: Outputs the payload data of the received long packet to the Video Pixel interface.

When the reception of a long packet is completed, the value indicated by WC and the data payload length are checked, a CRC check is performed, and the status is notified according to the check result.

1. If the data payload length is shorter than the value indicated by WC, set VCST(M).WCE to 1.
2. For a CRC error, set VCST(M).CRC to 1.

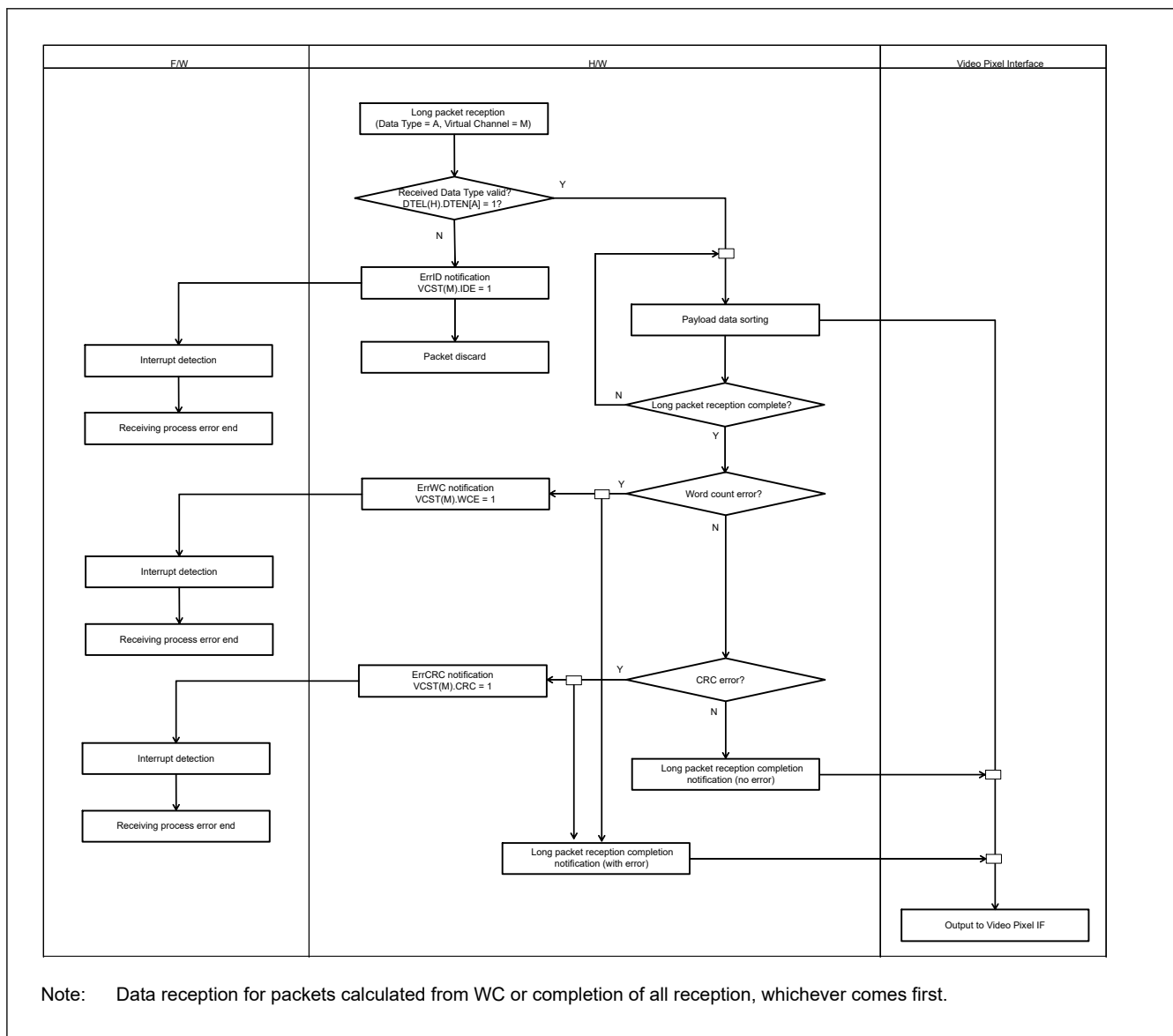


Figure 67.6 Long packet reception processing

67.4.3 Software Reset

By writing 1 to RTCT.VSRST, a reset of the signal output to the Video Pixel interface can be issued.

Figure 67.7 shows the flow when issuing a software reset by RTCT.VSRST.

Check the value of MCT3 before starting a software reset by RTCT.VSRST, and if MCT3.RXEN = 1, perform the receive operation stop processing described in (2) [Receiving operation stopped](#).

A software reset is executed by RTCT.VSRST in the receive operation stop processing flow in (2) [Receiving operation stopped](#).

Initiate a software reset by RTCT.VSRST in Figure 67.7 when (MCT3.RXEN = 0).

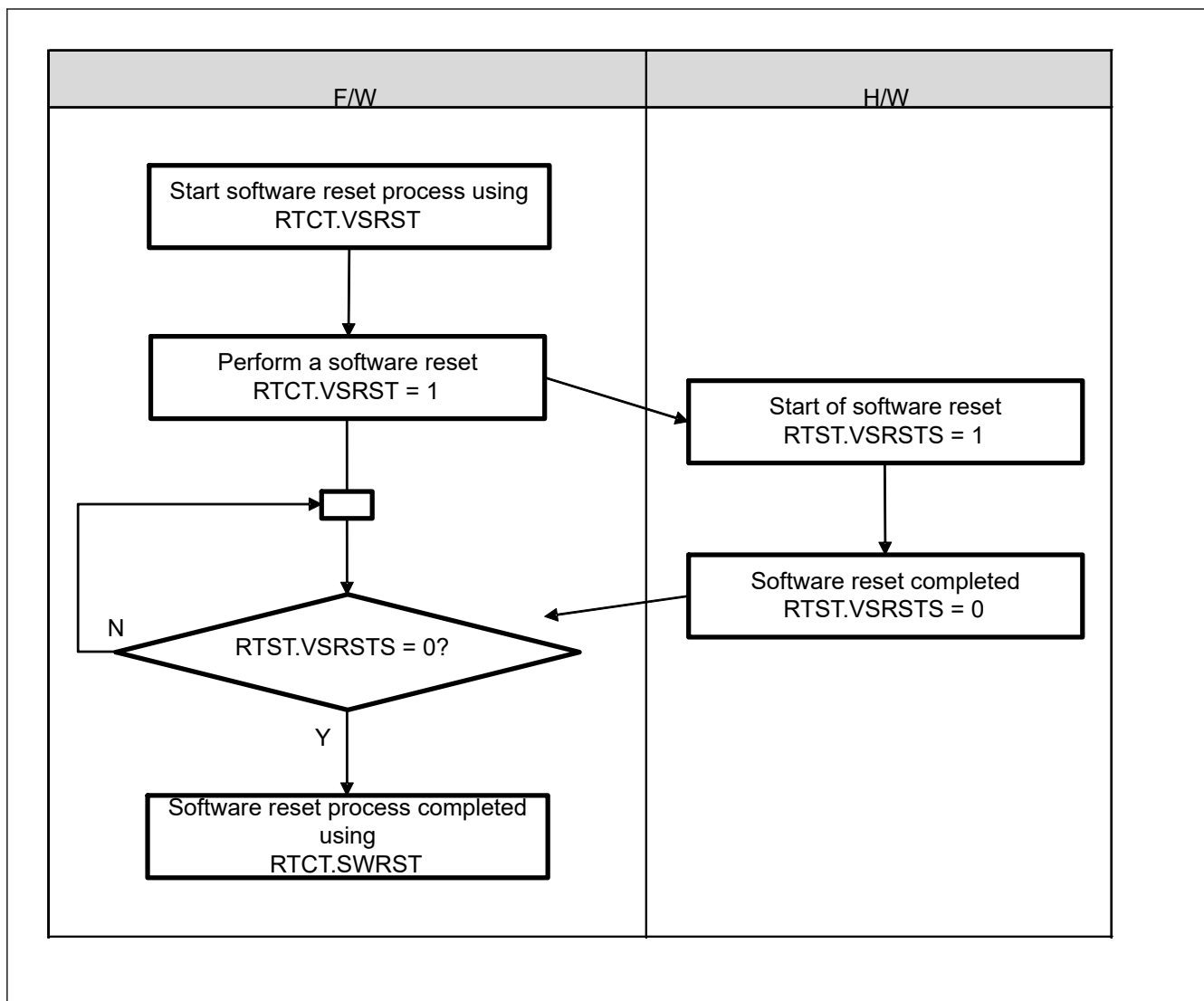


Figure 67.7 Software reset process (RTCT.VSRST)

67.5 Interrupt

Table 67.4 Interruption status and generation condition (1 of 4)

Event name	Interrupt status	Interrupt source register field	Interrupt enable register field	Generation condition
CSI0_RX	MIST.RXS	RXST.RACTDET	RXIE.RACTDETE	This module received a packet

Table 67.4 Interruption status and generation condition (2 of 4)

Event name	Interrupt status	Interrupt source register field	Interrupt enable register field	Generation condition
CSI0_DL	MIST.DL0S	DLST0.ESH	DLIE0.ESHE	ErrSotHS occurred on DL0
		DLST0.ESS	DLIE0.ESSE	ErrSotSynchS occurred on DL0
		DLST0.ECT	DLIE0.ECTE	ErrControl occurred on DL0
		DLST0.EES	DLIE0.EESE	ErrESC occurred on DL0
		DLST0.EUL	DLIE0.EULE	DL0 exited from the ULPS
		DLST0.RUL	DLIE0.RULE	DL0 transitioned to the ULPS
	MIST.DL1S	DLST1.ESH	DLIE1.ESHE	ErrSotHS occurred on DL1
		DLST1.ESS	DLIE1.ESSE	ErrSotSynchS occurred on DL1
		DLST1.ECT	DLIE1.ECTE	ErrControl occurred on DL1
		DLST1.EES	DLIE1.EESE	ErrESC occurred on DL1
		DLST1.EUL	DLIE1.EULE	DL1 exited from the ULPS
		DLST1.RUL	DLIE1.RULE	DL1 transitioned to the ULPS

Table 67.4 Interruption status and generation condition (3 of 4)

Event name	Interrupt status	Interrupt source register field	Interrupt enable register field	Generation condition
CSI0_VC	MIST.VC0S	VCST0.MLF	VCIE0.MLFE	Malformed packet was received on VC0
		VCST0.ECD	VCIE0.ECDE	ErrEccDouble occurred on VC0
		VCST0.CRC	VCIE0.CRCE	ErrCrc occurred on VC0
		VCST0.IDE	VCIE0.IDEE	ErrID occurred on VC0
		VCST0.WCE	VCIE0.WCEE	ErrWC occurred on VC0.
		VCST0.ECC	VCIE0.ECCE	ErrEccCorrected occurred on VC0
		VCST0.ECN	VCIE0.ECNE	ErrEccNoError occurred on VC0
		VCST0.FRS	VCIE0.FRSE	ErrFrameSync occurred on VC0
		VCST0.FRD	VCIE0.FRDE	ErrFrameData occurred on VC0
		VCST0.OVF	VCIE0.OVFE	Generic short packets on VC0 were discarded because GS FIFO overflowed
		VCST0.FSR	VCIE0.FSRE	Frame Start was received on VC0
		VCST0.FER	VCIE0.FERE	Frame End was received on VC0
		VCST0.LSR	VCIE0.LSRE	Line Start was received on VC0
		VCST0.LER	VCIE0.LERE	Line End was received on VC0
		VCST0.ETR	VCIE0.ETRE	EoTp was received on VC0.
	MIST.VCmS	VCSTm.MLF	VCIEm.MLFE	Malformed packet was received on VCm
		VCSTm.ECD	VCIEm.ECDE	ErrEccDouble occurred on VCm
		VCSTm.CRC	VCIEm.CRCE	ErrCrc occurred on VCm
		VCSTm.IDE	VCIEm.IDEE	ErrID occurred on VCm
		VCSTm.WCE	VCIEm.WCEE	ErrWC occurred on VCm
		VCSTm.ECC	VCIEm.ECCE	ErrEccCorrected occurred on VCm
		VCSTm.ECN	VCIEm.ECNE	ErrEccNoError occurred on VCm
		VCSTm.FRS	VCIEm.FRSE	ErrFrameSync occurred on VCm
		VCSTm.FRD	VCIEm.FRDE	ErrFrameData occurred on VCm
		VCSTm.OVF	VCIEm.OVFE	Generic short packets on VCm were discarded because GS FIFO overflowed
		VCSTm.FSR	VCIEm.FSRE	Frame Start was received on VCm
		VCSTm.FER	VCIEm.FERE	Frame End was received on VCm
CSI0_PM	MIST.PMS	PMST.DSX	PMIE.DSXE	All data lanes exited from the stop state
		PMST.DSN	PMIE.DSNE	All data lanes transitioned to the stop state
		PMST.CSX	PMIE.CSXE	The clock lane exited from the stop state
		PMST.CSN	PMIE.CSNE	The clock lane transitioned to the stop state
		PMST.DUX	PMIE.DUXE	All data lanes exited from the ULPS.
		PMST.DUN	PMIE.DUNE	All data lanes transitioned to the ULPS
		PMST.CUX	PMIE.CUXE	The clock lane exited from the ULPS
		PMST.CUN	PMIE.CUNE	The clock lane transitioned to the ULPS

Table 67.4 Interruption status and generation condition (4 of 4)

Event name	Interrupt status	Interrupt source register field	Interrupt enable register field	Generation condition
CSI0_GST	MIST.GSTS	GSST.GNE	GSIE.GNEE	GS FIFO is not empty
		GSST.GTH	GSIE.GTHE	Data stored in GS FIFO equals or exceeds the threshold
		GSST.GOV	GSIE.GOVE	GS FIFO overflowed

Note: The m value used in the table for CSI0_VC is 1 to 15.

67.6 Usage Notes

67.6.1 Power Gating Control or Software Standby Mode

Before transitioning to graphics power domain off or Software Standby mode, stop the operation as described in the following sections.

- [\(2\) Receiving operation stopped.](#)

After returning from graphics power domain on or Software Standby mode, resume operation as described in the following sections.

- [\(1\) Start of reception operation.](#)

67.6.2 Module-Stop Function Setting

MIPI CSI operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The MIPI CSI module is initially stopped after reset. Releasing the module-stop state enables access to the registers.